ITCS448 Data Warehousing and Business Intelligence (2/2025)
Project Phase 1 Data Warehouse Analysis and Design

Project Title: Automotive
Business Case: Toyota
Group Number: 4
Group Members:

No.	Student ID	First name and Lastname	Role	Responsibilities
1.	6688072	Kemjira Nugboon	Data Analyst	Identified Key Performance Indicators (KPIs), defined measures (facts), and analyzed source data characteristics.
2.	6688099	Nuttanan Reamprasert	Database Specialist	Developed the Data Dictionary, defined attribute types, and documented project benefits and limitations.
3.	6688129	Patsatraporn Thongdeesakul	Data Modeler	Defined the Grain of fact tables and designed the Star Schema diagrams for Sales, Service, and Inventory processes.
4.	6688168	Pornthita Sukprapaipat	Project Manager	Overlooked the project lifecycle, finalized the Executive Summary, and ensured alignment between project objectives and the final design.
5.	6688210	Thanaporn Pinsakul	Business Analyst	Conducted business requirements analysis, identified stakeholders, and defined key business questions.

Executive Summary

In today's highly competitive automotive industry, organizations must leverage data as a strategic asset to support informed operational and executive decision-making. Toyota Motor Corporation generates immense volumes of data through vehicle sales, dealership operations, inventory management, and after-sales services; however, these datasets are often siloed in disparate systems, hindering effective integration and comprehensive analysis.

To address these challenges, this project focuses on the analysis and design of a centralized dimensional data warehouse based on a Star Schema architecture. By integrating public Kaggle datasets with realistic synthetic data, the proposed system provides a unified analytical environment for stakeholders, including sales, supply chain, and service managers. The design is specifically engineered to monitor critical Key Performance Indicators (KPIs) such as regional sales trends, technician labor productivity, and inventory aging, which are vital for optimizing business performance.

This data warehouse supports cross-functional analysis of vehicle model performance and customer loyalty, empowering management to transition from reactive reporting to proactive strategic planning. While the project acknowledges challenges regarding data quality and integration complexity, the resulting framework establishes a scalable foundation for business intelligence. Ultimately, this design demonstrates how transforming raw automotive data into structured insights can significantly enhance operational efficiency, strengthen customer relationship management, and ensure Toyota's long-term sustainable growth in the digital era.

1. Introduction

1.1 Background of the Business Case

The automotive industry is highly competitive and continuously changing due to technological innovation, digital transformation, and evolving customer expectations. Companies in this industry must respond to global competition, sustainability trends, and increasing demand for data-driven decision-making. Toyota Motor Corporation is one of the world's leading automotive manufacturers, operating in more than 70 countries and producing a wide range of vehicles, including hybrid and electric cars. The company focuses on quality, innovation, and customer satisfaction to maintain its global competitiveness.

Toyota generates a large amount of data from various business activities such as vehicles sales, dealership operations, inventory management, customer information, and after sales services. However, this data is often stored in different systems across departments and regions, which makes it difficult to integrate and analyze effectively. As a result, decision-makers may face challenges in accessing timely and accurate insights. In today's digital era, organizations increasingly rely on business intelligence and data analytics to improve operational efficiency and support strategic planning. Therefore, implementing a data warehouse is essential to integrate data from multiple sources and provide a unified view of business performance. This project aims to design a data warehouse solution to support Toyota's business intelligence and decision-making processes.

1.2 Project Objectives

The main objective of this project is to design a data warehouse that supports business intelligence and improves decision-making for Toyota's automotive business. The objectives are divided into business and analytical perspectives as follows:

Business Objectives

- To improve visibility of sales performance across regions and dealerships.
- To support better inventory and supply chain management.
- To enhance customer relationship management and retention.
- To monitor after-sales service performance and warranty claims.
- To support strategic planning and business growth.

Analytical Objectives

- To identify key performance indicators (KPIs) related to sales, customers, and services.
- To analyze vehicle model performance and regional trends.
- To support customer behavior and repurchase analysis.
- To evaluate dealership performance and profitability.
- To enable data-driven decision making through dashboards and reports.

1.3 Project Scope and Assumptions

This project focuses on the analysis and design of a dimensional data warehouse for Toyota's automotive business. The scope includes business requirements analysis, source data analysis, dimensional modeling, and schema design. The proposed system aims to support analytical reporting, performance monitoring, and decision-making. However, the project does not include the full implementation of the system, real-time data integration, or deployment in a production environment.

The data used in this project is obtained from publicly available Kaggle datasets, including Toyota Used Cars and Car Sales datasets, along with synthetic datasets related to dealership, customers, service records, and inventory. These datasets are used to simulate real-world business scenarios due to privacy and data accessibility limitations.

Several assumptions are made in this project. First, the selected datasets represent typical automotive business operations. Second, synthetic data is assumed to reflect realistic customer and dealership behavior. Third, the data warehouse is designed for analytical and decision-support purposes rather than operational transaction processing. Finally, it is assumed that Toyota has sufficient infrastructure and resources to implement the proposed solution in the future.

2. Data Warehouse Analysis

2.1 Business Requirements Analysis

Stakeholders and Decision Makers

- Sales Managers: Monitor sales performance and track revenue targets.
- Dealership Managers: Evaluate dealership operations and overall profitability.
- Supply Chain Managers: Track inventory movement and manage stock shortages.
- Service Department Managers: Monitor service demand and track warranty-related issues.
- Executive Management: Focused on strategic planning and high-level performance evaluation.

Key Business Questions

- Which vehicle models generate the highest revenue per region?
- Which dealerships exhibit the best sales performance and profitability?
- How does inventory turnover vary across different regions and vehicle types?
- What is the average service revenue and frequency for each specific vehicle model?
- Which customer segments show the highest likelihood of repurchasing Toyota vehicles?

2.2 Analytical Requirements

Key Performance Indicators (KPIs)

1. Sales Performance Focuses on revenue generation and sales efficiency.

- Total Revenue: The sum of all sales transactions (sales_total) to monitor business growth.
- Average Sales Price (ASP): Calculated from sales_price to understand the average value per vehicle sold.
- Sales Volume: Total quantity to track the number of units moved in a specific period.
- Discount Variance: The difference between base_price (from Dim_Car) and sales_price to measure the impact of discounts on profit margins.
- Promotion Effectiveness: Analysis of sales performance categorized by promo_type (from Dim_Promotion) to evaluate marketing ROI.

2. Service & Maintenance Efficiency Focuses on workshop productivity and customer satisfaction.

- Service Revenue: Total income generated from maintenance and repairs (service_price).
- Labor Productivity: Measured by service_duration_hours to evaluate technician performance.

- Vehicle Downtime: Average downtime_days to track how long a customer's vehicle is out of service.
- Warranty Ratio: The percentage of service tasks flagged as warranty_flag (Yes/No) vs. paid services to manage operational costs.
- Service Throughput: Total number of service transactions (count of rows in Fact_Service) per dealership.

3. Inventory Management Focuses on stock health and availability.

- Inventory Asset Value: The total monetary value of vehicles currently in stock (inventory_value).
- Stock Availability Rate: The ratio of available_qty vs. total stock to ensure sales readiness.
- Inventory Aging: Monitoring how long a car_id stays in stock using date_id to identify slow-moving models.
- Stock Movement Analysis: Comparing stock_in and stock_out to determine which models have the highest demand and optimize replenishment.
- Sell-through Rate: Comparing units sold (stock_out) against units received (stock_in) to evaluate inventory turnover efficiency.

Measures (Facts)

1.Sales Measures

- sales_quantity: The number of vehicle units sold per transaction.
- sales_price: The actual price at which the vehicle was sold after negotiations.
- discount_amount: The total value of discounts or promotional price reductions applied.
- sales_total: The total revenue generated per transaction (sales_price *sales_quantity).

2.Service Measures

- service_price: Total service charge billed to the customer (service revenue).
- service_duration_hours: Total service time in hours used to measure technician productivity.
- downtime_days: Number of days the vehicle remained in service.
- discount_amount: Total discount applied to the service charge.

3.Inventory Measures

- stock_in: The number of units received into the warehouse from suppliers.
- stock_out: The number of units removed from inventory due to sales or transfers.
- stock_on_hand: The current balance of units available in physical inventory.
- inventory_value: The total monetary value of the current stock for financial reporting.

- reserved_qty: The number of vehicle units reserved by customers but not yet delivered.
- available_qty: The number of units actually available for sale, calculated as stock_on_hand - reserved_qty.

Dimensions

- **Dim_Date**: Facilitates temporal analysis, allowing the CEO to monitor sales trends, service peaks, and inventory turnover on a daily, monthly, quarterly, and yearly basis.
- **Dim_Car**: Enables performance evaluation by vehicle characteristics such as model, color, transmission type, fuel type, engine size, and price segment (based on base_price).
- **Dim_Dealership**: Allows for regional sales comparison, branch performance benchmarking, and geographical analysis across different locations and store sizes.
- **Dim_Customer**: Supports customer segmentation analysis, demographic evaluation (occupation, income band), and tracking of long-term loyalty or repurchase behavior.
- **Dim_Salesperson**: Evaluates individual sales performance and commission tracking based on employee position, status, and hire date.
- **Dim_Promotion**: Measures the impact of different marketing campaigns and discount types on sales volume and revenue.
- **Dim_PaymentMethod**: Analyzes customer payment preferences, including bank-specific installments and interest rate impacts on sales.
- **Dim_Technician**: Monitors staff efficiency in the service department by tracking service quality, technician rank, and task completion time.
- **Dim_ServiceType**: Categorizes maintenance tasks (e.g., routine check, engine repair) and assigns priority levels to optimize workshop scheduling.
- **Dim_Supplier**: Enables supplier performance tracking, focusing on procurement volume, regional sourcing, and lead time reliability.
- **Dim_Location**: Provides granular details of specific storage areas or warehouse bins within the dealership to optimize internal logistics.

2.3 Source Data Analysis

Our data is collected from two primary sources:

1. Toyota Dataset

- Source: Toyota Dataset from Kaggle
- Description: This dataset contains Toyota vehicle specifications and pricing information. It is primarily used to construct the Car entity in the OLTP database.
- Key Attributes (car):
    - car_id
    - model
    - year
    - price

    - transmission
    - fuel_type

This dataset supports vehicle-level analysis such as price comparison by model, year-based trend analysis, and transmission/fuel-type segmentation.

2. Car Sales Dataset

- Source: [Car Sales Dataset from Kaggle]
- Description: This dataset contains transactional car sales records including dealer and customer-related information. It is used to construct transactional and operational tables in the OLTP schema.
- Key Attributes (sales_transaction):
    - transaction_id
    - sale_date
    - car_id
    - quantity
    - total_amount
- Key Attributes (dealership):
    - dealer_id
    - dealer_name
    - region
- Key Attributes (customer):
    - customer_id
    - gender
    - customer_segment

This dataset supports sales performance analysis, dealership comparison, and customer behavior evaluation.

Additionally, our team generated three synthetic datasets for further analysis, including:

- dealership
- customer
- inventory

These datasets were created due to privacy constraints, as detailed operational and personal data were not publicly available. The synthetic data was generated based on realistic automotive business assumptions to simulate Toyota dealership operations and support dimensional modeling, ETL design, and business intelligence analysis in this project.

3. Data Warehouse Design

3.1 Dimensional Modeling

Grain Definition

- **Sales Fact:** One car sale transaction for one customer, on one date, at one dealership, handled by one salesperson, with one promotion and one payment method.
- **Service Fact:** One service transaction for one car, performed for one customer, on one date, at one dealership, handled by one technician, for one service type, with one payment method.
- **Inventory Fact:** One inventory record for one car, at one dealership, in one specific location, supplied by one supplier, on one date.

Fact Tables

Our model includes three primary fact tables to address key business questions:

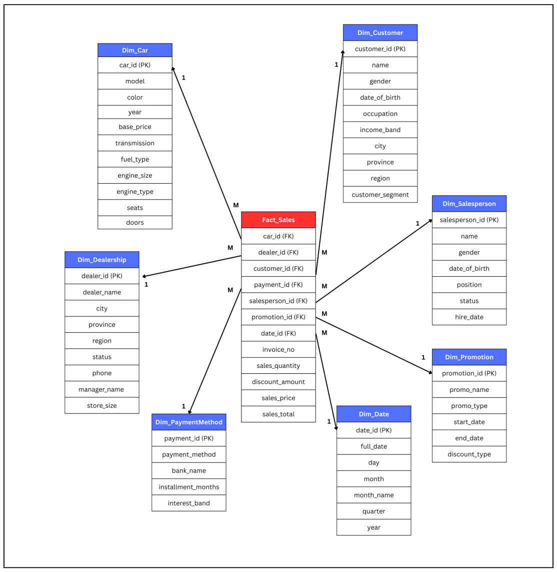
- **Fact_Sales:** invoice_no, sales_quantity, discount_amount, sales_price, sales_total
- **Fact_Service:** service_price, service_duration_hours, downtime_days, warranty_flag, discount_amount
- **Fact_Inventory:** stock_on_hand, stock_in, stock_out, reserved_qty, inventory_value, available_qty

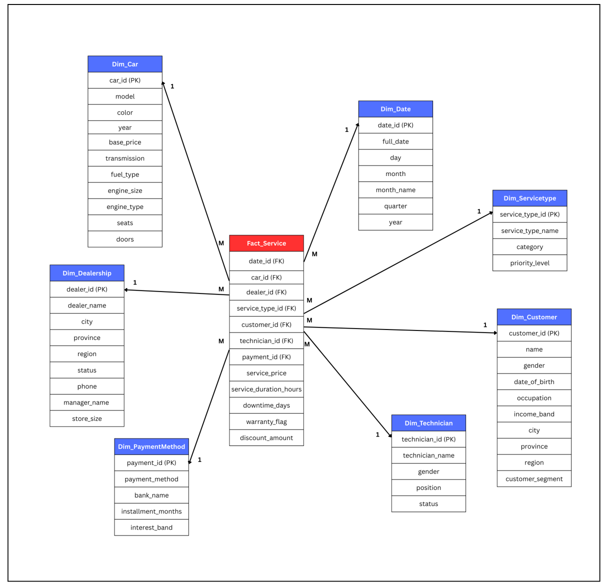
Dimension Tables

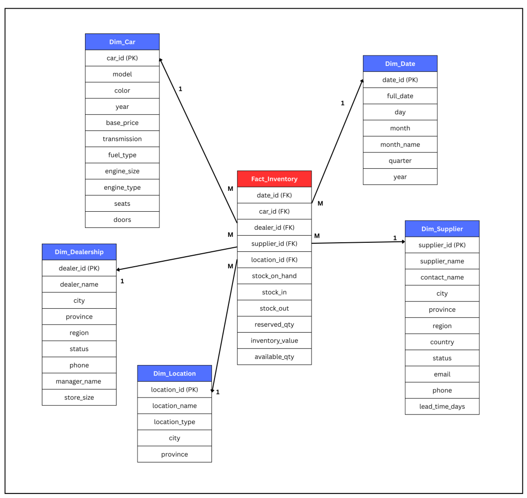
To provide contextual information for analysis, we utilize eleven main dimensions:

- **Dim_Date:** date_id, full_date, day, month, month_name, quarter, year
- **Dim_Car:** car_id, model, color, year, base_price, transmission, fuel_type, engine_size, engine_type, seats, doors
- **Dim_Dealership:** dealer_id, dealer_name, city, province, region, status, phone, manager_name, store_size
- **Dim_Customer:** customer_id, name, gender, date_of_birth, occupation, income_band, city, province, region, customer_segment
- **Dim_Salesperson:** salesperson_id, name, gender, date_of_birth, position, status, hire_date
- **Dim_PaymentMethod:** payment_id, payment_method, bank_name, installment_months, interest_band
- **Dim_Promotion:** promotion_id, promo_name, promo_type, start_date, end_date, discount_type
- **Dim_Technician**: technician_id, technician_name, gender, position, status
- **Dim_ServiceType**: service_type_id, service_type_name, category, priority_level
- **Dim_Supplier**: supplier_id, supplier_name, contact_name, city, province, region, country, status, lead_time_days
- **Dim_Location**: location_id, location_name, location_type, city, province

3.2 Data Warehouse Schema



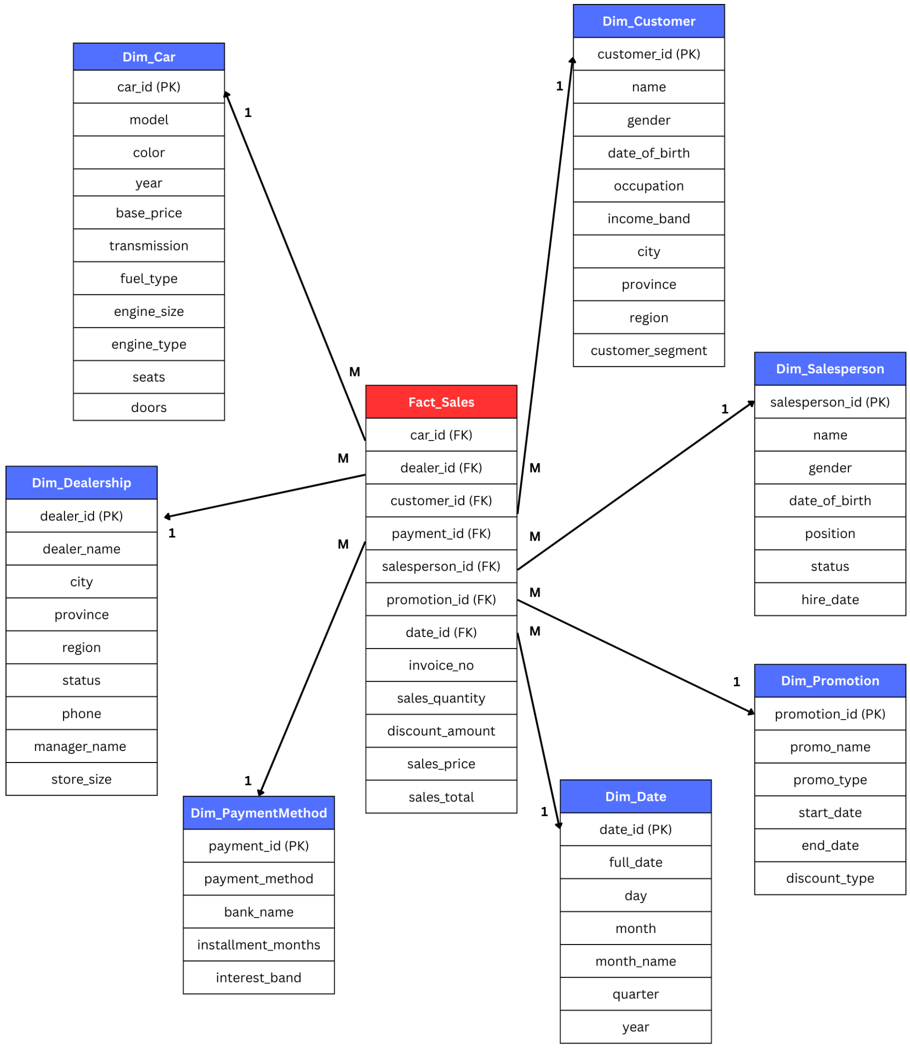




Link for full star schema

This overall star schema represents an integrated automotive dealership (TOYOTA) data warehouse consisting of three fact tables: Fact_Sales, Fact_Service, and Fact_Inventory. It connects core business processes vehicle sales, after-sales service, and inventory management through shared dimensions such as Car, Date, and Dealership. The model enables comprehensive analysis of revenue, service performance, stock levels, customer behavior, operational efficiency, and supplier management, supporting strategic decision-making across the entire dealership operation.

1) Fact_Sales Star Schema



This Star Schema is designed to analyze Toyota's sales performance by capturing granular transactional data, including units sold, prices, and discounts. By integrating dimensions like Dim_Date, Dim_Car, and Dim_Customer, the system provides a view of regional revenue trends and vehicle model performance. This design transforms raw sales records into actionable insights to support strategic decision-making and profitability tracking.

Grain:
One car sale transaction for one customer, on one date, at one dealership, handled by one salesperson, with one promotion and one payment method.

Measures:

- sales_quantity
- discount_amount

- sales_price (unit price)
- sales_total (final revenue after discount)

Degenerate Dimension:

- invoice_no (transaction identifier stored directly in the fact table)

Connected Dimensions and Analytical Purpose:

- **Dim_Date**
  Supports sales trend analysis over time (daily, monthly, quarterly, yearly).

- **Dim_Car**
  Enables performance comparison by vehicle model, transmission type, fuel type, and manufacturing year.

- **Dim_Customer**
  Supports customer segmentation, demographic analysis, and repurchase behavior tracking.

- **Dim_Dealership**
  Allows regional sales comparison and dealership performance evaluation.

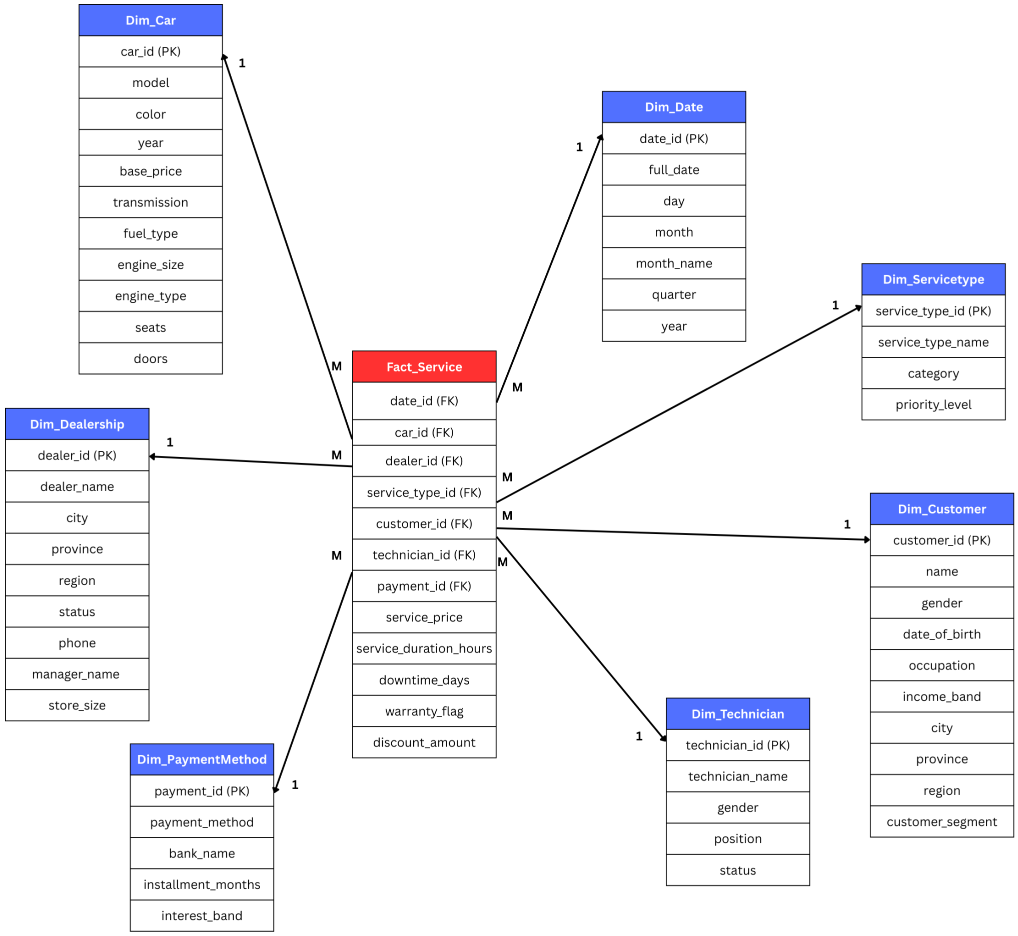
- **Dim_Salesperson**
  Enables salesperson performance analysis, commission tracking, and productivity comparison.

- **Dim_Promotion**
  Supports evaluation of promotion effectiveness and discount impact on sales performance.

- **Dim_PaymentMethod**
  Allows analysis of customer payment preferences (cash, financing, leasing, etc.) and revenue distribution by payment type.

The overall business purpose of this schema is to analyze vehicle sales performance across products, customers, locations, and time to support revenue growth, profitability improvement, and strategic decision-making.

2) Fact_Service Star Schema



The Fact_Service Star Schema monitors the operational efficiency and financial health of the after-sales department by tracking service revenue, labor duration, and downtime. By analyzing these metrics against Dim_Technician and Dim_ServiceType, management can evaluate technician productivity and service category demand. This framework enables Toyota to optimize resource allocation and enhance customer retention through improved service quality.

Grain:
One service transaction for one car, performed for one customer, on one date, at one dealership, handled by one technician, for one service type, with one payment method.

Measures:

- service_price (unit price)
- service_duration_hours
- downtime_days
- discount_amount

Degenerate Dimension:

- warranty_flag (Service status indicator used to distinguish between warranty and paid services)

Connected Dimensions and Analytical Purpose:

- **Dim_Date**

  Supports service trend analysis over time (daily workload, monthly trends, seasonal patterns, yearly growth).

- **Dim_Car**

  Enables analysis of service frequency by car model, fuel type, transmission type, and manufacturing year.

- **Dim_Customer**

  Supports customer retention analysis, service frequency tracking, and after-sales behavior insights.

- **Dim_Dealership**

  Allows comparison of service revenue and operational performance across dealership locations.

- **Dim_Technician**

  Enables technician productivity analysis, performance comparison, and labor efficiency evaluation.

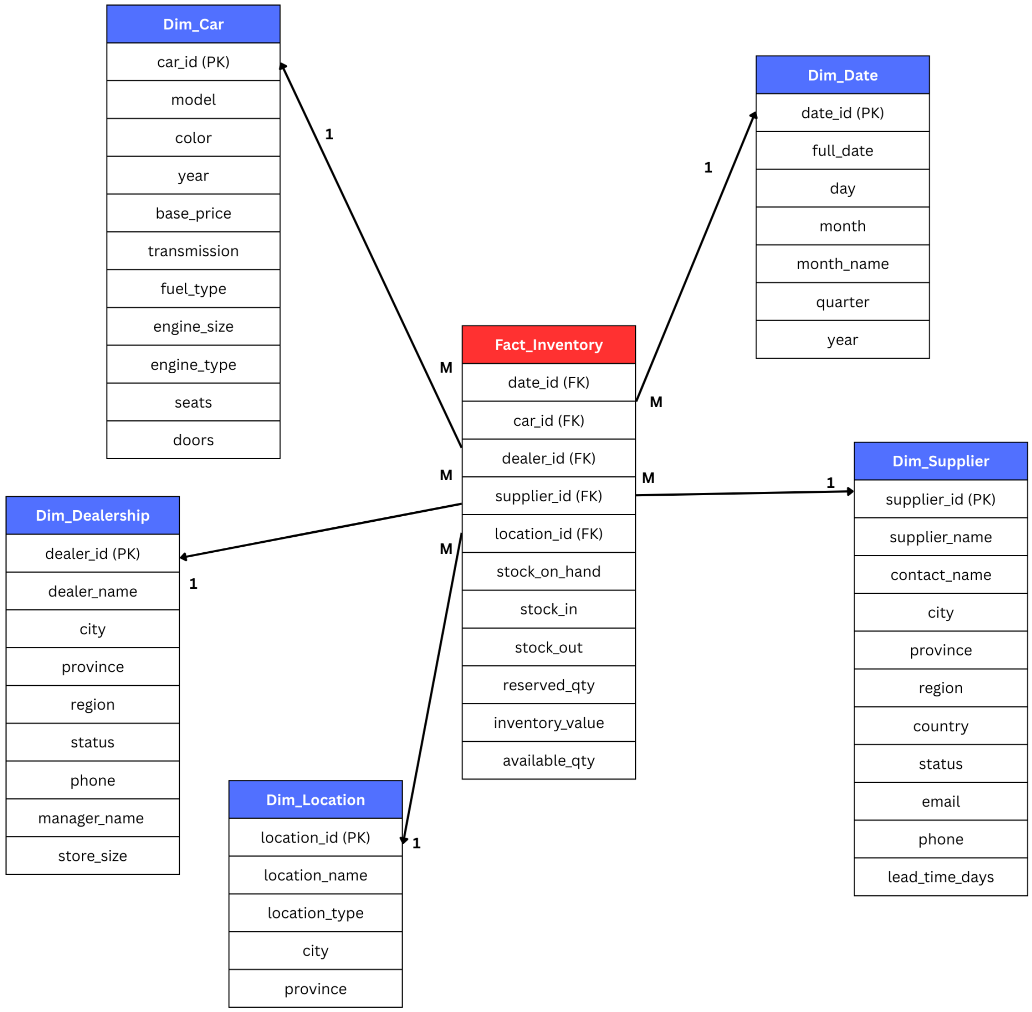
- **Dim_ServiceType**

  Supports analysis of service category performance (maintenance, repair, warranty, inspection, etc.).

- **Dim_PaymentMethod**
  Allows analysis of customer payment preferences and revenue distribution by payment type.

This schema enables analysis of service revenue, technician performance, service trends over time, dealership efficiency, customer retention, and payment behavior.

3) Fact_Inventory Star Schema



This Star Schema serves as a periodic snapshot for tracking stock levels and optimizing supply chain logistics across Toyota's dealership network. The central Fact_Inventory table monitors stock movement including stock-in, stock-out, and available quantities to maintain high visibility of asset value. This approach allows for the identification of slow-moving inventory and the evaluation of supplier reliability to minimize carrying costs.

Grain:

One inventory record for one car, at one dealership, in one specific location, supplied by one supplier, on one date.

Measures:

- stock_on_hand
- stock_in
- stock_out

- reserved_qty
- available_qty
- inventory_value

Connected Dimensions and Analytical Purpose:

- **Dim_Date**
  Supports inventory trend analysis over time and stock aging evaluation.

- **Dim_Car**
  Enables analysis of stock levels by car model, type, and year.

- **Dim_Dealership**
  Allows comparison of inventory levels across dealership branches.

- **Dim_Location**
  Supports warehouse or storage-level inventory tracking.

- **Dim_Supplier**
  Enables supplier performance evaluation and supply chain analysis.

The scheme is to track vehicle stock levels across dealerships over time to optimize inventory control and cost management.

3.3 Data Dictionary

Fact Tables

1) Fact_Sales

Table	Attribute	Type	Format	Key	Reference	Description
Fact_Sales	date_id	INT	11	FK	Dim_Date (date_id)	Links to Dim_Date
Fact_Sales	car_id	INT	11	FK	Dim_Car (car_id)	Links to Dim_Car
Fact_Sales	dealer_id	INT	11	FK	Dim_Dealership (dealer_id)	Links to Dim_Dealership
Fact_Sales	customer_id	INT	11	FK	Dim_Customer (customer_id)	Links to Dim_Customer
Fact_Sales	payment_id	INT	11	FK	Dim_PaymentMethod (payment_id)	Links to Dim_PaymentMethod
Fact_Sales	salesperson_id	INT	11	FK	Dim_Salesperson (salesperson_id)	Links to Dim_Salesperson
Fact_Sales	promotion_id	INT	11	FK	Dim_Promotion (promotion_id)	Links to Dim_Promotion

Fact_Sales	invoice_no	VARCHAR	30	-	-	Unique transaction number (degenerate dimension)
Fact_Sales	sales_quantity	INT	11	-	-	Units sold
Fact_Sales	discount_amount	DECIMAL	10,2	-	-	Total discount applied to the sale
Fact_Sales	sales_price	DECIMAL	10,2	-	-	Unit selling price after discount
Fact_Sales	sales_total	DECIMAL	12,2	-	-	Total revenue (sales_price * sales_quantity)

2) Fact_Service

Table	Attribute	Type	Format	Key	Reference	Description
Fact_Service	date_id	INT	11	FK	Dim_Date (date_id)	Links to Dim_Date
Fact_Service	car_id	INT	11	FK	Dim_Car (car_id)	Links to Dim_Car
Fact_Service	dealer_id	INT	11	FK	Dim_Dealership (dealer_id)	Links to Dim_Dealership
Fact_Service	service_type_id	INT	11	FK	Dim_ServiceType	Links to Dim_ServiceType
Fact_Service	customer_id	INT	11	FK	Dim_Customer (customer_id)	Links to Dim_Customer
Fact_Service	technician_id	INT	11	FK	Dim_Technician (technician_id)	Links to Dim_Technician
Fact_Service	payment_id	INT	11	FK	Dim_PaymentMethod (payment_id)	Links to Dim_PaymentMethod
Fact_Service	service_price	DECIMAL	10,2	-	-	Total service charge before discount
Fact_Service	service_duration_hours	DECIMAL	5,2	-	-	Total service time in hours
Fact_Service	downtime_days	INT	11	-	-	Number of days the car stayed in service
Fact_Service	warranty_flag	CHAR(1)	Y, N	-	-	Indicates whether the service is under

						warranty (Yes/No)
Fact_Service	discount_amount	DECIMAL	10,2	-	-	Total discount applied to the service charge

3) Fact_Inventory

Table	Attribute	Type	Format	Key	Reference	Description
Fact_Inventory	date_id	INT	11	FK	Dim_Date (date_id)	Links to Dim_Date
Fact_Inventory	car_id	INT	11	FK	Dim_Car(car_id)	Links to Dim_Car
Fact_Inventory	dealer_id	INT	11	FK	Dim_Dealership (dealer_id)	Links to Dim_Dealership
Fact_Inventory	supplier_id	INT	11	FK	Dim_Supplier	Links to Dim_Supplier
Fact_Inventory	location_id	INT	11	FK	Dim_Location (location_id)	Links to Dim_Location
Fact_Inventory	stock_on_hand	INT	11	-	-	Current stock balance on that date (units)
Fact_Inventory	stock_in	INT	11	-	-	Units received into inventory on that date
Fact_Inventory	stock_out	INT	11	-	-	Units removed from inventory on that date (sales/transfers)
Fact_Inventory	reserved_qty	INT	11	-	-	Units reserved/not available for sale
Fact_Inventory	inventory_value	DECIMAL	12,2	-	-	Total value of inventory on that date
Fact_Inventory	available_qty	INT	11	-	-	Units available for sale (stock_on_hand – reserved_qty)

Dimension Tables

1) Dim_Date

Table	Attribute	Type	Format	Key	Reference	Description
Dim_Date	date_id	INT	11	PK	-	Surrogate key for date
Dim_Date	full_date	DATE	YYYY-MM-DD	-	-	Actual calendar date
Dim_Date	day	INT	2	-	-	Day of month
Dim_Date	month	INT	2	-	-	Month number
Dim_Date	month_name	VARCHAR	20	-	-	Month name
Dim_Date	quarter	INT	1	-	-	Quarter number
Dim_Date	Year	INT	4	-	-	Year

2) Dim_Car

Table	Attribute	Type	Format	Key	Reference	Description
Dim_Car	car_id	INT	11	PK	-	Surrogate key for car
Dim_Car	model	VARCHAR	50	-	-	Car model
Dim_Car	color	VARCHAR	50	-	-	Color of car
Dim_Car	year	INT	4	-	-	Manufacture year
Dim_Car	base_price	DECIMAL	10,2	-	-	Standard price before discount
Dim_Car	transmission	VARCHAR	20	-	-	Transmission type
Dim_Car	fuel_type	VARCHAR	20	-	-	Fuel type
Dim_Car	engine_size	DECIMAL	3,1	-	-	Engine capacity
Dim_Car	engine_type	VARCHAR	50	-	-	Engine configuration
Dim_Car	seats	INT		-	-	Number of seats
Dim_Car	doors	INT		-	-	Number of doors

3) Dim_Dealership

Table	Attribute	Type	Format	Key	Reference	Description
Dim_Dealership	dealer_id	INT	11	PK	-	Surrogate key for dealership
Dim_Dealership	dealer_name	VARCHAR	100	-	-	Dealership name
Dim_Dealership	city	VARCHAR	50	-	-	City of dealership
Dim_Dealership	province	VARCHAR	50	-	-	Province/state of dealership
Dim_Dealership	region	VARCHAR	50	-	-	Geographic region
Dim_Dealership	status	VARCHAR	30	-	-	Operational status
Dim_Dealership	phone	VARCHAR	20	-	-	Dealership contact number
Dim_Dealership	manager_name	VARCHAR	50	-	-	Dealership manager name
Dim_Dealership	store_size	VARCHAR	10	-	-	Small, Medium, Large

4) Dim_Customer

Table	Attribute	Type	Format	Key	Reference	Description
Dim_Customer	customer_id	INT	11	PK	-	Surrogate key for customer
Dim_Customer	name	VARCHAR	50	-	-	Full name of customer
Dim_Customer	gender	VARCHAR	50	-	-	Male, Female, Other
Dim_Customer	date_of_birth	DATE	YYYY-MM-DD	-	-	Customer date of birth (used to derive age group)
Dim_Customer	occupation	VARCHAR	50	-	-	Customer's occupation
Dim_Customer	income_band	VARCHAR	20	-	-	Income category (Low, Medium, High)
Dim_Customer	city	VARCHAR	20	-	-	Customer's city
Dim_Customer	province	VARCHAR	50	-	-	Customer's province
Dim_Customer	region	VARCHAR	50	-	-	Customer's region
Dim_Customer	customer_segment	VARCHAR	50	-	-	Retail, Corporate, etc.

5) Dim_Salesperson

Table	Attribute	Type	Format	Key	Reference	Description
Dim_Salesperson	salesperson_id	INT	11	PK	-	Surrogate key for salesperson
Dim_Salesperson	name	VARCHAR	50	-	-	Full name of the salesperson
Dim_Salesperson	gender	VARCHAR	50	-	-	Gender of the salesperson
Dim_Salesperson	date_of_birth	DATE	YYYY-MM-DD	-	-	Birth date of the salesperson
Dim_Salesperson	position	VARCHAR	30	-	-	Job position or role
Dim_Salesperson	status	VARCHAR	20	-	-	Employment status
Dim_Salesperson	hire_date	DATE	YYYY-MM-DD	-	-	Date the salesperson was hired

6) Dim_PaymentMethod

Table	Attribute	Type	Format	Key	Reference	Description
Dim_PaymentMethod	payment_id	INT	11	PK	-	Surrogate key for payment
Dim_PaymentMethod	payment_method	VARCHAR	20	-	-	Type of payment (Cash, Credit Card, Bank Loan, Leasing)
Dim_PaymentMethod	bank_name	VARCHAR	50	-	-	Name of the bank or financial institution
Dim_PaymentMethod	installment_months	INT	2	-	-	Number of months for installment payments
Dim_PaymentMethod	interest_band	VARCHAR	10	-	-	Interest rate category (Low, Medium, High)

7) Dim_ServiceType

Table	Attribute	Type	Format	Key	Reference	Description
Dim_ServiceType	service_type_id	INT	11	PK	-	Surrogate key for service type
Dim_ServiceType	service_type_name	VARCHAR	50	-	-	Name of the service (Oil Change, Engine Repair)
Dim_ServiceType	category	VARCHAR	20	-	-	Service category (Maintenance, Repair)
Dim_ServiceType	priority_level	VARCHAR	10	-	-	Priority classification (Low, Medium, High)

8) Dim_Technician

Table	Attribute	Type	Format	Key	Reference	Description
Dim_Technician	technician_id	INT	11	PK	-	Surrogate key for technician
Dim_Technician	technician_name	VARCHAR	50	-	-	Full name of the technician
Dim_Technician	gender	VARCHAR	50	-	-	Gender of the technician
Dim_Technician	position	VARCHAR	50	-	-	Job role or specialization
Dim_Technician	status	VARCHAR	20	-	-	Employment status (Active, On Leave)

9) Dim_Location

Table	Attribute	Type	Format	Key	Reference	Description
Dim_Location	location_id	INT	11	PK	-	Surrogate key for location
Dim_Location	location_name	VARCHAR	50	-	-	Name of the location
Dim_Location	location_type	VARCHAR	50	-	-	Type of location
Dim_Location	city	VARCHAR	50	-	-	City where the location is situated
Dim_Location	province	VARCHAR	50	-	-	Province of the location

10) Dim_Supplier

Table	Attribute	Type	Format	Key	Reference	Description
Dim_Supplier	supplier_id	INT	11	PK	-	Surrogate key for supplier
Dim_Supplier	supplier_name	VARCHAR	50	-	-	Name of the supplier company
Dim_Supplier	contact_name	VARCHAR	50	-	-	Main contact person
Dim_Supplier	city	VARCHAR	50	-	-	City of the supplier
Dim_Supplier	province	VARCHAR	50	-	-	Province of the supplier
Dim_Supplier	region	VARCHAR	50	-	-	Geographic region
Dim_Supplier	country	VARCHAR	50	-	-	Country of the supplier
Dim_Supplier	status	VARCHAR	20	-	-	Supplier status
Dim_Supplier	email	VARCHAR	50	-	-	Supplier contact email
Dim_Supplier	phone	VARCHAR	20	-	-	Supplier contact phone number
Dim_Supplier	lead_time_days	INT	2	-	-	Average delivery lead time in days

11) Dim_Promotion

Table	Attribute	Type	Format	Key	Reference	Description
Dim_Promotion	promotion_id	INT	11	PK	-	Surrogate key for promotion
Dim_Promotion	promo_name	VARCHAR	50	-	-	Promotion campaign name
Dim_Promotion	promo_type	VARCHAR	30	-	-	Type of promotion (Seasonal, Clearance, etc.)
Dim_Promotion	start_date	DATE	YYYY-MM-DD	-	-	Promotion start date
Dim_Promotion	end_date	DATE	YYYY-MM-DD	-	-	Promotion end date
Dim_Promotion	discount_type	VARCHAR	30	-	-	Discount method (Percent, Fixed amount, etc.)

4. Discussion

4.1 Benefits

1. **Integrated Business Analysis**
   The data warehouse integrates sales, service, and inventory data into a single analytical environment. This integration allows management to evaluate dealership performance holistically, compare revenue streams, and understand the relationship between vehicle sales and after-sales services without relying on separate operational systems.
2. **KPI Monitoring and Trend Analysis**
   By organizing data into fact and dimension tables, the system enables efficient monitoring of key performance indicators such as total revenue, service revenue, and stock levels. The Time dimension supports flexible aggregation (daily, monthly, quarterly, yearly), allowing users to identify trends, seasonality, and performance patterns.
3. **Scalability and Flexibility**
   The dimensional model is scalable and adaptable. Additional dimensions (e.g., promotion campaigns or financing options) and new fact tables (e.g., warranty claims) can be incorporated with minimal structural changes, ensuring long-term usability.

4.2 Challenges

1. **Data Quality Issues**
   Public datasets often contain missing values, inconsistent formats, and limited attribute details. Significant data cleaning and transformation efforts are required before loading the data into the warehouse, increasing ETL complexity.
2. **Use of Synthetic Data**
   Synthetic datasets were generated due to privacy limitations. Although they simulate dealership operations, they may not fully represent real-world complexity. Analytical outcomes may therefore differ from real operational scenarios.
3. **ETL Integration Complexity**
   Integrating multiple datasets with different schemas and granularities requires careful mapping and validation. Maintaining consistent keys and aligning transaction-level data with dimension tables demands precise design and testing.

4.3 Limitations

1. **Limited Real Operational Data**
   The absence of actual dealership operational records restricts analytical depth. Real-world datasets would provide richer financial, marketing, and service details.
2. **No Real-Time Data Processing**
   The current design supports batch ETL processing only. It does not incorporate real-time or streaming data pipelines, limiting immediate dashboard updates.
3. **Simplified Customer Dimension**
   Customer attributes are limited to basic demographics, restricting advanced analytics such as customer lifetime value analysis, churn prediction, or personalized marketing segmentation.

5. Conclusion

This project successfully designed and documented a dimensional data warehouse model to support automotive business analysis for Toyota. By translating business requirements into analytical requirements, the team developed a structured star schema that integrates sales, service, and inventory data into a unified analytical environment. The defined grain of the fact tables ensures consistency and accuracy in aggregation, while shared dimension tables enable flexible cross-functional analysis. The resulting design supports key performance indicators such as total revenue, service revenue trends, and stock monitoring across dealerships and vehicle models.

Through this project, the team gained practical experience in applying data warehousing concepts, including business requirement analysis, KPI identification, dimensional modeling, fact and dimension table design, and schema normalization decisions. The project also strengthened understanding of ETL considerations, data integration challenges, and the importance of aligning technical design with managerial decision-making needs.

For future improvements, the data warehouse can be enhanced by integrating real dealership operational data, incorporating more detailed customer demographics and behavioral attributes, and implementing real-time data pipelines. Additionally, advanced analytics techniques such as predictive modeling for sales forecasting, inventory optimization algorithms, and customer segmentation models can further increase the strategic value of the system.

Overall, the project demonstrates how data warehousing and business intelligence techniques can transform raw automotive data into structured insights that support informed managerial decisions and long-term business growth.

References

- Toyota Motor Corporation. (2023). Annual Report 2023. Retrieved from https://global.toyota/en/ir/library/annual/
- Kaggle. (2024). *Toyota Dataset*. Retrieved from https://www.kaggle.com/datasets/kingarther1527/toyota
- Kaggle. (2024). *Car Sales Data*. Retrieved from https://www.kaggle.com/datasets/suraj520/car-sales-data